MRA Assumptions Exploration

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Model building

Linear regression relies on several assumptions. In this project I explore what happens when those assumptions are violated, and more specifically **which estimated quantities change the most** when each violation is introduced in isolation. I focus on:

1. **Linearity**: the conditional mean of the outcome is a linear function of the predictors.
2. **Normality of errors**: the residuals are normally distributed around zero.
3. **Homoscedasticity**: the error variance is constant across fitted values.
4. **Independence of errors**: residuals are uncorrelated across observations.

To study this, I simulate a "perfect" model first and then alter one assumption at a time. For each scenario, I compare the resulting:

- intercept
- regression coefficients for working memory and IQ
- standard errors
- residual standard error
- R^2

I also keep one representative dataset from each scenario to visualize how the assumption is violated.

The data generating model is:

$$\text{math\_test}_i = 22 + 0.2\,\text{IQ}_i + 1.1\,\text{WM}_i + \varepsilon_i, \qquad i = 1, \dots, n.$$

with predictors generated as

$$\text{IQ}_i \overset{iid}{\sim} \mathcal{N}(100, 15^2), \qquad \text{WM}_i \overset{iid}{\sim} \mathcal{N}(26, 8^2),$$

and under the benchmark model,

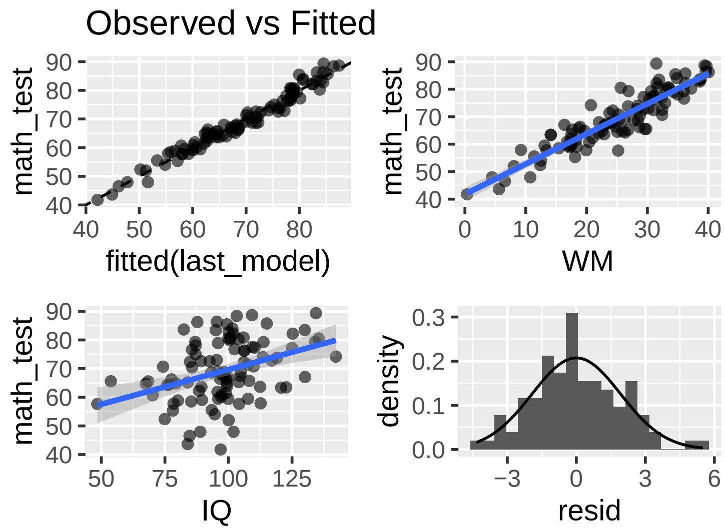
$$\varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, 2^2).$$

So under the ideal case,

$$\text{math\_test}_i \mid (\text{IQ}_i, \text{WM}_i) \sim \mathcal{N}\left(22 + 0.2\,\text{IQ}_i + 1.1\,\text{WM}_i, 2^2\right).$$

Perfect model

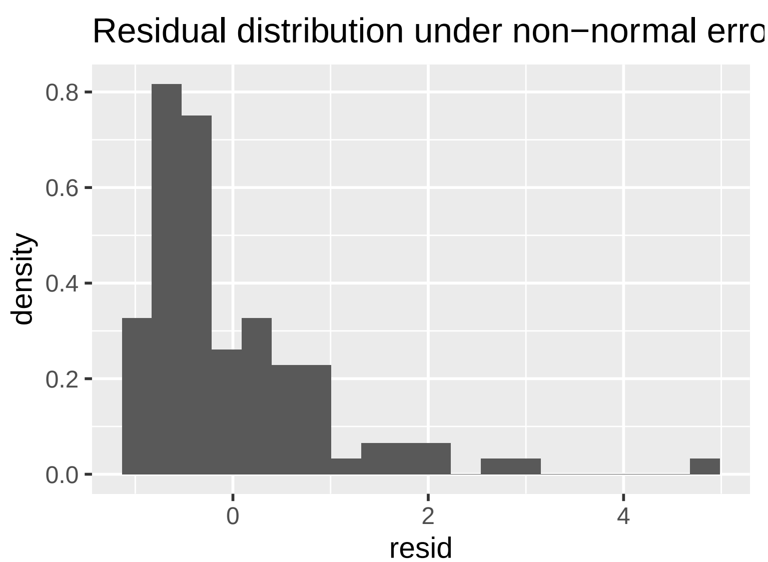
Visualizations for the perfect model



The plots do not show violations of the key assumptions of multiple regression. The dependent variable and the independent variables have a linear relationship, the errors are approximately normal, and have constant variance across the fitted values.

Violation of Normality

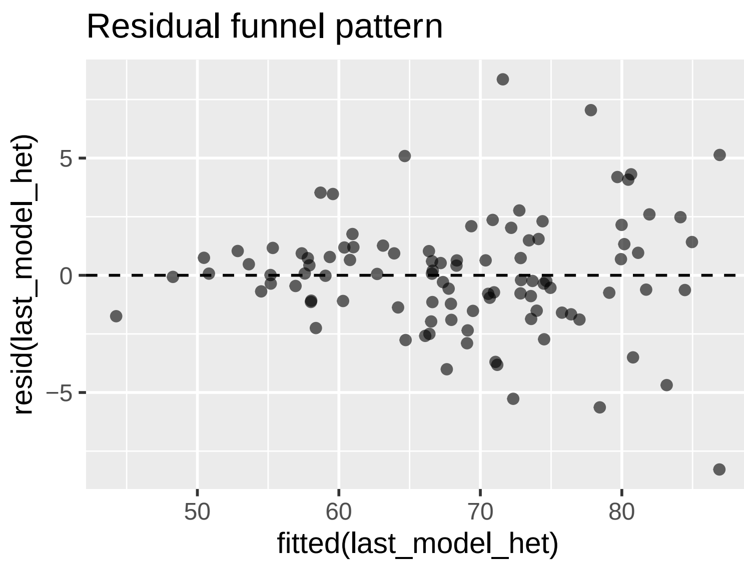
To violate the normality assumption, I generate skewed errors using an exponential distribution.



Even though the residual distribution is clearly skewed, the regression coefficients remain close to the true values. This illustrates that **normality mainly affects inference rather than the coefficient estimates themselves**.

Violation of Homoscedasticity

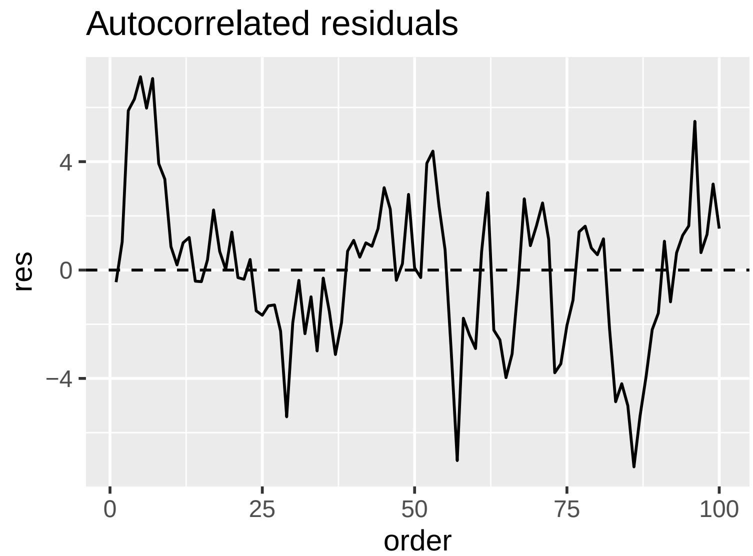
Now the variance of the errors increases with working memory.



The funnel shape indicates increasing variance. Again, the coefficients remain largely unbiased, but **standard errors are affected because the model assumes constant variance**.

Violation of Independence

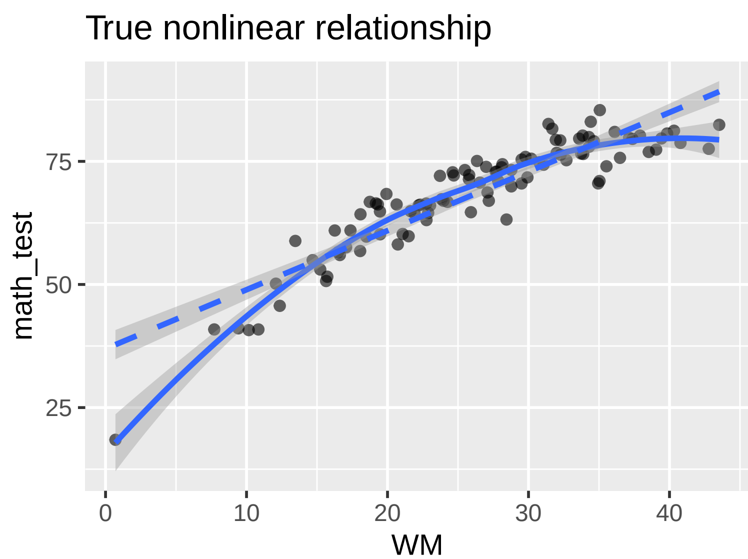
Autocorrelation is introduced by making each residual depend on the previous residual.



Residuals now show a clear pattern over observations, violating independence.

Violation of Linearity

Finally, I introduce a quadratic relationship with working memory.



Here the linear regression tries to approximate a curved relationship with a straight line. This leads to **biased slope estimates**, making this assumption violation particularly problematic.

Overall Conclusion

The simulation illustrates that not all regression assumptions affect the model in the same way.

Violations of **normality**, **homoscedasticity**, and **independence** mainly affect the **uncertainty estimates** of the model. The estimated regression coefficients themselves remain relatively stable, but the standard errors and therefore statistical inference become unreliable.

In contrast, violations of **linearity** directly distort the estimated relationship between predictors and outcome. Because the model attempts to fit a straight line to a nonlinear pattern, the regression coefficients themselves become biased.

This simulation therefore highlights that the most consequential violation for interpreting regression coefficients is typically **misspecification of the mean structure**, while the other assumptions primarily influence the reliability of inference.